GROUP PROJECT PROPOSAL

FixFlow AI

An intelligent maintenance management platform for organizations

CSE327.3 | Group 9

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THE PROBLEM

Where informal maintenance fails

Lost requests

Calls and chat messages are never properly recorded.

Hard to prioritize

A broken chair looks the same as an electrical hazard.

Wrong technician

Managers match skills to jobs by hand.

No transparency

Reporters never learn whether anyone is working on it.

Repeated failures

The same equipment keeps failing and nobody sees the pattern.

No performance data

No view of request volumes, problem locations or repair times.

THE SOLUTION

One workflow from report to closure

1	2	3	4	5	6	7
Report	AI analysis	Duplicate check	Manager review	Assign technician	Repair and update	User verifies, ticket closes

Example: “AC running but not cooling” becomes HVAC, Medium severity, 91% confidence, and Ticket #FF-1024 is created.

Status path: Reported → Under Review → Assigned → In Progress → Resolved → Verified → Closed. If the user rejects the fix, the ticket is Reopened.

THE AI COMPONENT

Four AI assists, managers decide

AI gives recommendations. Managers can accept or override every one.

1 Problem classification

“Water leaking from the ceiling in Room 205” is classified as Plumbing.

2 Severity recommendation

Low, Medium, High or Critical, with a reason. Sparks from a power socket is Critical.

3 Duplicate detection

Three reports about the Conference Room 2 AC point to one existing open ticket.

4 Technician recommendation

Matches skills, availability, current workload and location.

USERS AND FEATURES

Four roles, one platform

Requester

- Report an issue with a photo
- Track status
- Get notifications
- Verify the repair

Manager

- Review AI suggestions
- Set priority
- Assign technicians
- View analytics

Technician

- See assigned jobs
- Update status
- Add repair notes
- Upload evidence

Administrator

- Manage users
- Manage locations and categories
- Organization settings
- System reports

ARCHITECTURE AND STACK

Three layers plus external services

React.js frontend

Forms, dashboards and request tracking for all four roles, on desktop and mobile browsers.

Node.js + Express backend

REST APIs, authentication, assignment, notifications and analytics.

Python + FastAPI AI service

Classification, severity, duplicate similarity and technician recommendation.

PostgreSQL / MySQL database

Users, locations, requests, assignments, status history and feedback.

External services

Email, SMS and push notifications, plus cloud storage for photos.

SCOPE

A focused, achievable MVP

In scope

- Login and role-based access
- Request form with location and photo
- AI category and severity
- Duplicate request detection
- Technician assignment
- Status tracking and notifications
- Repair history and user verification
- Manager dashboard and analytics

REQUIREMENTS

Functional requirements

ID	Type	Description
FR1	Functional	The system shall authenticate users according to their assigned roles.
FR2	Functional	The system shall allow users to submit maintenance requests with location, description and optional images.
FR3	Functional	The system shall use an AI component to suggest the category and severity of a maintenance request.
FR4	Functional	The system shall identify potentially duplicate unresolved maintenance requests.
FR5	Functional	The system shall allow managers to assign maintenance requests to technicians.
FR6	Functional	The system shall allow technicians to update request status and add repair information.
FR7	Functional	The system shall allow requesters to verify completed maintenance requests and reopen them if the problem remains unresolved.

REQUIREMENTS

Non-functional requirements

ID	Type	Description
NFR1	Non-functional	The system shall normally respond to standard user requests within 2 seconds under the defined prototype load.
NFR2	Non-functional	The system shall enforce role-based authorization for protected functions.
NFR3	Non-functional	The system shall maintain an audit history of important request status changes.
NFR4	Non-functional	The system shall provide a responsive interface for desktop and mobile browsers.

RISKS

Known risks and how we handle them

Risk	Our response
AI misclassifies category or severity	AI only recommends; managers can override
Limited training data	Start with common categories and a curated dataset
False duplicate matches	Matches are suggestions, not automatic merges
Unauthorized access	Role-based authentication and authorization
Server downtime	Database backups and error handling
Large project scope	Build the MVP first, then add advanced features

NOVELTY

What makes FixFlow AI different

Products like UpKeep prove there is demand. FixFlow AI is a focused, affordable platform for small and medium-sized organizations.

Explainable AI

“Critical because the report indicates an electrical hazard.”

Human in the loop

Managers approve or override every AI decision.

Affordable

Built for organizations that do not need a huge enterprise system.

Flexible

Configurable for hotels, offices, schools, apartments, hospitals and factories.

BUSINESS MODEL

Monthly and yearly subscription plans

Subscription: each organization pays a fixed fee, billed monthly or yearly, in BDT.

Starter

BDT **2,000** / month

BDT 25,500 / year (save 15%)

Small offices, shops, apartments

Business

BDT **6,000** / month

BDT 66,000 / year (save 15%)

Schools, hotels, clinics

Enterprise

Custom quote

Monthly or yearly contract

Hospitals, factories, chains

Also: one-time setup and training fee · SMS alerts as a paid add-on

PLAN AND TEAM

Eight weeks, three members

8-week plan

1. Requirements, architecture and UI/UX design
2. Database, backend, frontend and AI module
3. Integration and testing
4. Deployment, documentation and final presentation

Responsibilities

Member 1: Backend and system integration

Member 2: UI/UX design and frontend

Member 3: AI, database and QA

Shared: requirements, UML, GitHub, testing, documentation and the final presentation.

FIXFLOW AI

Thank you

Questions?

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